Lecture 13: Why Are People Different?: Differences

Introduction to Psychology · Yale University, Spring 2007 · Professor Paul Bloom

Study guide based on the lecture transcript. Source: https://oyc.yale.edu/psychology/psyc-110/lecture-13 (CC BY-NC-SA 3.0).

Before you start

Until now the course has studied what all people share. This lecture asks why people are different, with a focus on two main differences: personality and intelligence. Professor Bloom explains how these are measured and what research says about genes and environment, including findings that many people find shocking.

Learning goals

After this lecture you should be able to:

- Explain the difference between reliability and validity, with an example.
- Describe the Big Five personality traits and the evidence for them.
- Explain Spearman's "G" and how IQ scores are calculated.
- Explain what heritability means and how twin and adoption studies separate genes from environment.
- Describe the two surprising findings of behavioral genetics and explain why parent-child correlations are hard to interpret.

Key ideas

1. Personality, Intelligence and Determining Difference

- Personality is a person's style of dealing with the world, especially other people. It is a stable trait across situations and time: being angry once after someone hits you is not personality; walking around angry all the time is.
- A good test needs reliability (it gives the same result at different times, like a good bathroom scale) and validity (it measures what it is supposed to measure). An intelligence test based on your date of birth would be reliable but not valid.
- The Rorschach inkblot test is widely used, even in child custody cases, but Bloom says it is "as useless as throwing dice."
- Earlier theories differed on the number of personality dimensions. Eysenck first proposed two (introverted-extroverted and neurotic-stable) and later added a third; Cattell proposed sixteen.

2. Measuring Personality with "The Big Five"

- Today psychologists agree on "The Big Five." Remember them with the word OCEAN: openness, conscientiousness, extroversion, agreeableness, and neuroticism.

- Evidence: the traits are stable over time, especially after age thirty. Different observers (a roommate, a professor, a mother) tend to agree about a person.
- They predict real behavior. Conscientiousness relates to faithfulness to a spouse, openness to changing jobs, and extroverts look people in the eye more.
- People who rated characters from "The Simpsons" agreed that Ned Flanders and Marge were very agreeable, and Mr. Burns and Nelson were not.

3. Defining and Measuring Intelligence

- In a survey of 1,000 experts, almost everyone said intelligence involves abstract reasoning, problem solving, and the capacity to acquire knowledge.
- Spearman proposed "S" (your ability on a specific test) and "G" (a general intelligence you bring to all tests). It is like sport: someone good at running and swimming is probably good at climbing too.
- On modern tests such as the Wechsler test, 100 is the average. 68% of people score between 85 and 115, and 95% score between 70 and 130.
- The Bell Curve, by Herrnstein and Murray, argued that IQ matters greatly for success in society. Expert reports written in response agreed that IQ is strongly related to many educational, occupational, economic and social outcomes.
- Part of this may be a self-fulfilling prophecy, because society uses IQ-like tests (the SAT, the GRE) to choose students. But IQ also relates to mental speed and memory span.

4. Question and Answer on Personality and Intelligence

Test makers catch people who lie with unrealistic "catch questions" (such as "I have never done anything I am ashamed of"). Personality and IQ can change, but where you stand now is significantly related to where you will be later.

5. The Roles of Genes and Environment in Explaining Human Differences

- Genes and environment explain differences between people, not one person's trait. Height is quite heritable. The number of legs is not, because people with fewer legs usually lost them in accidents, even though having legs is genetic.
- Shared environment is what children raised in the same home share, such as their parents. Non-shared environment is everything else, such as a snowball that hit you at age five.
- Behavioral genetics compares monozygotic (identical) twins, who share 100% of their genes, dizygotic twins, who share 50% like ordinary siblings, and adopted siblings, who have no special genetic overlap. The "gold standard" is identical twins raised apart. They are surprisingly alike, even in their brain scans.
- First surprising finding: there is high heritability for almost everything, including intelligence, personality, happiness, and political orientation.
- There are reliable differences in average IQ scores among American racial groups, and these groups are partly socially constructed. But high heritability within a group does not mean that differences between groups are genetic. In Richard Lewontin's example, seeds grow in two plots of land, and one plot gets much more fertilizer. Within each plot, growth differences are genetic. The difference between the plots comes from the fertilizer.

6. Genes, Environment and Intelligence

- Bloom gives two reasons from the textbook that group differences are at least largely environmental. First, IQ differences match socially defined groups better than genetically defined groups. Second, the Flynn effect: people have been getting smarter without genetic change. An average person in 1980 would score 120 on the 1950 test. Nobody knows why.
- Second, more surprising finding: almost everything that is not genetic comes from non-shared environment. Shared environment counts for little or nothing. The IQ correlation of genetically unrelated adults raised in the same family is about zero.
- So do parents matter? In The Nurture Assumption, Judith Harris argued that parents matter less, and peers more, than we think. Bloom says parents have a huge effect, but "at the moment of conception."
- Children do resemble their parents, but this can come from shared genes, or the child can affect the parent. A study claimed family meals keep teenagers away from drugs; Bloom called it "the worst study ever done." Perhaps good kids choose to come to meals, good families do both, or age explains it: twelve-year-olds eat with their family more and use fewer drugs than seventeen-year-olds.

Key vocabulary

Term	Meaning in simple English	Example from the lecture
personality	a person's stable style of dealing with the world and other people	Walking around angry all the time
reliability	a test gives the same result at different times	A bathroom scale that gives the same number
validity	a test measures what it is supposed to measure	A date-of-birth intelligence test is not valid
The Big Five (OCEAN)	five main dimensions of personality	Ned Flanders is high in agreeableness
G (general intelligence)	the general ability people bring to all intellectual tests	"I think he is high 'G.'"
IQ score	a result on an intelligence test; 100 is average	95% of people score between 70 and 130
self-fulfilling prophecy	something becomes true because society makes it important	Yale uses the SAT to choose students
heritability	how much of the differences between people is due to genes	Height is quite heritable; number of legs is not
shared environment	what children raised in the same home share	Having the same parents
non-shared environment	all other experiences	A snowball hitting you at age five

Term	Meaning in simple English	Example from the lecture
monozygotic / dizygotic twins	twins with 100% / 50% of the same genes	Identical twins raised apart are the "gold standard"
Flynn effect	people getting smarter over generations	The average person in 1980 would score 120 on the 1950 test

Check your understanding

1. What is the difference between reliability and validity? Use Bloom's date-of-birth test as an example.
2. What do the letters of OCEAN stand for? Give one piece of evidence that the Big Five are real.
3. Explain Spearman's "G" using Bloom's sports example.
4. A friend gets an IQ score of 115. What does this tell you, based on how IQ tests are scored?
5. Why is the number of legs a good example to explain heritability?
6. Why are identical twins raised apart called the "gold standard"?
7. What is the second, more surprising finding of behavioral genetics?
8. A study finds that teenagers who eat family meals use fewer drugs. Give two other explanations.

Answer guide

1. Reliability means the same result each time; validity means measuring what the test should measure. A date-of-birth test is reliable, but it has nothing to do with intelligence.
2. Openness, conscientiousness, extroversion, agreeableness, neuroticism. The traits are stable over time, observers agree, and they predict behavior.
3. People who are good at one sport tend to be good at others. In the same way, beyond specific skills ("S"), people have a general ability ("G").
4. The average is 100, and 68% of people score between 85 and 115. So 115 is above average, at the top of that middle range.
5. Having legs is genetic, but differences in the number of legs usually come from accidents. Heritability is about differences between people.
6. They have the same genes but different environments, so their similarities suggest the effect of genes.
7. Almost everything that is not genetic comes from non-shared environment. Shared environment, including how parents raise children, counts for little or nothing.
8. Good kids may choose to come home for meals (reversed direction), or good families may produce both.

English practice

Useful phrases from the lecture

- "I want to be clear." - used before an important point that people often misunderstand.
- "To put it a different way, ..." - says the same idea again in other words.
- "It's equally possible it's the other way around." - says that cause and effect may be reversed.
- "The proof is sort of in the pudding." - an idiom: you judge something by its results.

Speaking task (with your AI tutor)

- Explain Lewontin's seed example in 4-6 sentences. Use: *heritability*, *within a group*, *between groups*.

Writing task

- Bloom's reading response question was "Explain the Flynn effect." Write one paragraph (80-120 words): say what the Flynn effect is, why IQ tests hide it, and what it shows about genes and environment.

Study with your tutor

Things to say to your tutor in Live conversation:

- "Let's review Lecture 13. Ask me questions about reliability and validity, one at a time."
- "Quiz me on the key vocabulary of Lecture 13."

- "I will explain heritability and twin studies in my own words. Check my understanding and my English."

Readings from the original course

- Gray, Peter. Psychology (5th edition), pp. 49-61, 362-380, 423-444, 538-557
- Herrnstein, Richard J. and Charles Murray. "The Bell Curve: Intelligence and Class Structure in American Life." In The Norton Psychology Reader. Edited by Gary Marcus. New York: W. W. Norton & Company, 2006. pp. 188-198
- Harris, Judith Rich. "The Nurture Assumptions: Why Children Turn Out the Way They Do." In The Norton Psychology Reader. Edited by Gary Marcus. New York: W. W. Norton & Company, 2006. pp. 291-303
- Gladwell, Malcolm. "Personality Plus." In The Norton Psychology Reader. Edited by Gary Marcus. New York: W. W. Norton & Company, 2006. pp. 304-316

My notes